Twelfth Grade Tile Task

Standard S.CP.1

12

1. **a)** Outcomes of a spinner with 1 equal sections — 1
   b) Outcomes of a spinner with 7 equal sections — 7
   c) Outcomes of a fair die? — 6

2. **a)** Outcomes of flipping two coins? — 4
   b) Outcomes of a spinner with 5 equal sections — 5
   c) Outcomes of a spinner with 3 equal sections — 3

3. **a)** Outcomes of a spinner with 8 equal sections — 8
   b) Outcomes of a spinner with 9 equal sections — 9
   c) Outcomes of a spinner with 2 equal sections — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.2

12

1. a) Outcomes where two dice sum to 7? **6**

b) Outcomes where a die is 5? **1**

c) Outcomes where a die shows 7? **0**

2. a) Outcomes of a spinner with 9 equal sections? **9**

b) Outcomes of a spinner with 7 equal sections? **7**

c) Outcomes where two dice sum to 8? **5**

3. a) Outcomes where two dice sum to 4? **3**

b) Outcomes where a die is greater than 2? **4**

c) Outcomes of a spinner with 8 equal sections? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.3

12

1. **a)** Favorable 1 out of 7 equally likely outcomes: numerator — 1
   b) Favorable 8 out of 9 equally likely outcomes: numerator — 8
   c) Favorable 4 out of 6 equally likely outcomes: numerator — 4

2. **a)** Favorable 3 out of 5 equally likely outcomes: numerator — 3
   b) Favorable 2 out of 5 equally likely outcomes: numerator — 2
   c) Favorable 7 out of 7 equally likely outcomes: numerator — 7

3. **a)** Favorable 9 out of 9 equally likely outcomes: numerator — 9
   b) Favorable 6 out of 8 equally likely outcomes: numerator — 6
   c) Favorable 5 out of 7 equally likely outcomes: numerator — 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.4

12

1. a) Two-way table: 5 yes and 0 no. Total? **5**

b) Two-way table: 7 yes and 2 no. Total? **9**

c) Two-way table: 7 yes and 1 no. Total? **8**

2. a) Two-way table: 1 yes and 1 no. Total? **2**

b) Two-way table: 2 yes and 1 no. Total? **3**

c) Two-way table: 4 yes and 3 no. Total? **7**

3. a) Two-way table: 0 yes and 1 no. Total? **1**

b) Two-way table: 4 yes and 2 no. Total? **6**

c) Two-way table: 0 yes and 4 no. Total? **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.5

12

1. a) 4 of 10 students are in both groups. How many? **4**

b) 3 of 10 students are in both groups. How many? **3**

c) 8 of 10 students are in both groups. How many? **8**

2. a) 1 of 10 students are in both groups. How many? **1**

b) 5 of 10 students are in both groups. How many? **5**

c) 0 of 10 students are in both groups. How many? **0**

3. a) 7 of 10 students are in both groups. How many? **7**

b) 2 of 10 students are in both groups. How many? **2**

c) 9 of 10 students are in both groups. How many? **9**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.6

12

1. **a)** Chance of an event is 7 out of 10. Numerator? **7**
   b) Chance of an event is 0 out of 10. Numerator? **0**
   c) Chance of an event is 6 out of 10. Numerator? **6**

2. **a)** Chance of an event is 9 out of 10. Numerator? **9**
   b) Chance of an event is 1 out of 10. Numerator? **1**
   c) Chance of an event is 5 out of 10. Numerator? **5**

3. **a)** Chance of an event is 8 out of 10. Numerator? **8**
   b) Chance of an event is 3 out of 10. Numerator? **3**
   c) Chance of an event is 4 out of 10. Numerator? **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.CP.7

12

1. **a)** 4 of 10 have the trait. How many have it? **4**

   b) 3 of 10 have the trait. How many have it? **3**

   c) 5 of 10 have the trait. How many have it? **5**

2. **a)** 7 of 10 have the trait. How many have it? **7**

   b) 2 of 10 have the trait. How many have it? **2**

   c) 9 of 10 have the trait. How many have it? **9**

3. **a)** 6 of 10 have the trait. How many have it? **6**

   b) 8 of 10 have the trait. How many have it? **8**

   c) 0 of 10 have the trait. How many have it? **0**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.1

12

1. **a)** A random sample has 7 people. 7

   b) A random sample has 4 people. 4

   c) A random sample has 1 people. 1

2. **a)** A random sample has 6 people. 6

   b) A random sample has 2 people. 2

   c) A random sample has 9 people. 9

3. **a)** A random sample has 3 people. 3

   b) A random sample has 8 people. 8

   c) A random sample has 5 people. 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.2

12

1. a) Simulation: heads in 0 of 10 trials. **0**

b) Simulation: heads in 6 of 10 trials. **6**

c) Simulation: heads in 9 of 10 trials. **9**

2. a) Simulation: heads in 5 of 10 trials. **5**

b) Simulation: heads in 8 of 10 trials. **8**

c) Simulation: heads in 3 of 10 trials. **3**

3. a) Simulation: heads in 7 of 10 trials. **7**

b) Simulation: heads in 1 of 10 trials. **1**

c) Simulation: heads in 4 of 10 trials. **4**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.3

12

1. **a)** Survey sample size of 6 people — 6

   b) Survey sample size of 5 people — 5

   c) Survey sample size of 3 people — 3

2. **a)** Survey sample size of 8 people — 8

   b) Survey sample size of 7 people — 7

   c) Survey sample size of 1 people — 1

3. **a)** Survey sample size of 4 people — 4

   b) Survey sample size of 9 people — 9

   c) Survey sample size of 2 people — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.4

12

1. **a)** In a sample of 10, 3 said yes. How many said yes? **3**
   b) In a sample of 10, 7 said yes. How many said yes? **7**
   c) In a sample of 10, 8 said yes. How many said yes? **8**

2. **a)** In a sample of 10, 1 said yes. How many said yes? **1**
   b) In a sample of 10, 2 said yes. How many said yes? **2**
   c) In a sample of 10, 0 said yes. How many said yes? **0**

3. **a)** In a sample of 10, 9 said yes. How many said yes? **9**
   b) In a sample of 10, 6 said yes. How many said yes? **6**
   c) In a sample of 10, 5 said yes. How many said yes? **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.5

12

1. a) Group A mean 7, group B mean 5: difference? **2**

b) Group A mean 9, group B mean 5: difference? **4**

c) Group A mean 6, group B mean 1: difference? **5**

2. a) Group A mean 4, group B mean 3: difference? **1**

b) Group A mean 9, group B mean 2: difference? **7**

c) Group A mean 9, group B mean 0: difference? **9**

3. a) Group A mean 8, group B mean 5: difference? **3**

b) Group A mean 8, group B mean 2: difference? **6**

c) Group A mean 8, group B mean 0: difference? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.IC.6

12

1. **a)** Groups compared in a nine-group experiment? **9**
   b) Groups compared in a four-group experiment? **4**
   c) Groups compared in a three-group experiment? **3**

2. **a)** Groups compared in a five-group experiment? **5**
   b) Groups compared in a six-group experiment? **6**
   c) Groups compared in a one-group experiment? **1**

3. **a)** Groups compared in a seven-group experiment? **7**
   b) Groups compared in a two-group experiment? **2**
   c) Groups compared in an eight-group experiment? **8**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.ID.3

12

1. **a)** Median of 3, 6, 10 — 6

   b) Median of 8, 6, 9 — 8

   c) Median of 2, 8, 4 — 4

2. **a)** Median of 1, 10, 9 — 9

   b) Median of 4, 10, 7 — 7

   c) Median of 2, 6, 0 — 2

3. **a)** Median of 0, 6, 3 — 3

   b) Median of 2, 5, 7 — 5

   c) Median of 8, 0, 1 — 1

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.ID.4

12

1. **a)** Standard deviations in the 95% rule? 2

   b) Mean of a normal curve centered at 5? 5

   c) Mean of a normal curve centered at 9? 9

2. **a)** Mean of a normal curve centered at 6? 6

   b) Standard deviations in the 68% rule? 1

   c) Mean of a normal curve centered at 7? 7

3. **a)** Mean of a normal curve centered at 4? 4

   b) Mean of a normal curve centered at 8? 8

   c) Standard deviations in the 99.7% rule? 3

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

TILE TASKS: TWELFTH GRADE

Twelfth Grade Tile Task

Standard S.MD.1

12

1. **a)** Expected value of a game that pays 8 with probability 1 — **8**
   b) Expected value of a game that pays 5 with probability 1 — **5**
   c) Expected value of a game that pays 1 with probability 1 — **1**

2. **a)** Expected value of a game that pays 4 with probability 1 — **4**
   b) Expected value of a game that pays 3 with probability 1 — **3**
   c) Expected value of a game that pays 0 with probability 1 — **0**

3. **a)** Expected value of a game that pays 6 with probability 1 — **6**
   b) Expected value of a game that pays 2 with probability 1 — **2**
   c) Expected value of a game that pays 7 with probability 1 — **7**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.MD.2

12

1. **a)** Weighted average of 9 (weight 1) and 7 (weight 1) — 8
   b) Weighted average of 0 (weight 1) and 6 (weight 1) — 3
   c) Weighted average of 1 (weight 1) and 1 (weight 1) — 1

2. **a)** Weighted average of 9 (weight 1) and 9 (weight 1) — 9
   b) Weighted average of 7 (weight 1) and 1 (weight 1) — 4
   c) Weighted average of 4 (weight 1) and 0 (weight 1) — 2

3. **a)** Weighted average of 6 (weight 1) and 8 (weight 1) — 7
   b) Weighted average of 7 (weight 1) and 5 (weight 1) — 6
   c) Weighted average of 7 (weight 1) and 3 (weight 1) — 5

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.MD.3

12

1. **a)** Outcomes of a spinner with 1 sections: count — **1**
   b) Outcomes of a spinner with 8 sections: count — **8**
   c) Outcomes of a spinner with 3 sections: count — **3**

2. **a)** Outcomes of a spinner with 4 sections: count — **4**
   b) Outcomes of a spinner with 6 sections: count — **6**
   c) Outcomes of two coins with exactly one head — **2**

3. **a)** Outcomes of a spinner with 7 sections: count — **7**
   b) Outcomes of a spinner with 9 sections: count — **9**
   c) Outcomes of a spinner with 5 sections: count — **5**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.MD.4

12

1. **a)** Expected payoff of a fair game paying 1 — 1
   b) Expected payoff of a fair game paying 8 — 8
   c) Expected payoff of a fair game paying 7 — 7

2. **a)** Expected payoff of a fair game paying 4 — 4
   b) Expected payoff of a fair game paying 9 — 9
   c) Expected payoff of a fair game paying 5 — 5

3. **a)** Expected payoff of a fair game paying 3 — 3
   b) Expected payoff of a fair game paying 6 — 6
   c) Expected payoff of a fair game paying 2 — 2

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9

Twelfth Grade Tile Task

Standard S.MD.6

12

1. **a)** Fair price of a game paying 9 with probability 1 — **9**
   b) Fair price of a game paying 6 with probability 1 — **6**
   c) Fair price of a game paying 2 with probability 1 — **2**

2. **a)** Fair price of a game paying 7 with probability 1 — **7**
   b) Fair price of a game paying 8 with probability 1 — **8**
   c) Fair price of a game paying 0 with probability 1 — **0**

3. **a)** Fair price of a game paying 4 with probability 1 — **4**
   b) Fair price of a game paying 5 with probability 1 — **5**
   c) Fair price of a game paying 3 with probability 1 — **3**

Tile box: cut out the tiles and move them to your answer boxes

0	1	2	3	4	5	6	7	8	9